Krylov Complexity in the FRW Diamond

Extending Heller–Papalini–Schuhmann to Non-Stationary Cosmology via Conformal Pullback

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Companion note to: "The Modular Shadow Conjecture" [Letters in Mathematical Physics, submitted] and to: "Type II_∞ for the gravitational algebra in radiation-dominated FRW" [4]

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Abstract

We extend the analysis of Heller, Papalini, and Schuhmann [1] — which establishes a quantitative match between Krylov spread complexity in double-scaled SYK (DSSYK) and the complexity-equals-volume proposal in sine-dilaton gravity at finite temperatures — to the FRW causal diamond. The algebraic foundation is provided by the type-II_∞ classification of the FRW diamond algebra [4]: for the conformally coupled massless scalar in radiation- or matter-dominated FRW, the crossed-product algebra $\mathcal{A}(\mathcal{O})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ is type II_∞, with the modular automorphism group unitarily intertwined with the Hislop–Longo flow on the conformally identified Minkowski diamond. The KMS state at $\beta = 2\pi$ (conformal pullback of Bisognano–Wichmann) gives a modular Hamiltonian with continuous unbounded spectrum, enabling a Lanczos recursion. We conjecture [CONJECTURE] that the Krylov complexity growth rate in the FRW diamond satisfies $\lambda_K^{\mathrm{FRW}} = 2\pi/\beta = 1$ in natural units, saturating the Maldacena–Shenker–Stanford (MSS) chaos bound, as a structural consequence of the type-II_∞ classification and the KMS condition. The central bridge $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$ is flagged explicitly as unproven; the note records supporting arguments, identifies the obstructions, and proposes a programme of ~ 4 weeks of computation to either substantiate or refute the conjecture.

Contents

1 Background and Motivation **2**
- 1.1 Krylov complexity and the MSS chaos bound 2
- 1.2 Heller–Papalini–Schuhmann: DSSYK and holographic Krylov 2
- 1.3 Modular shadow and MSS saturation (ECI sec:mss-shadow) 3

2 The Heller–Papalini–Schuhmann Setting and Its Scope **4**
- 2.1 DSSYK and sine-dilaton gravity 4
- 2.2 What is AdS/static-specific vs. what can transport to FRW 4

3 FRW Diamond Setup **5**
- 3.1 The metric and the comoving observer 5
- 3.2 Conformally coupled scalar and the type-II_∞ theorem 5

3.3 The FRW modular Hamiltonian and its KMS state 6
3.4 Scope and honest limitations 6

4 Krylov Complexity in the FRW Diamond **7**
4.1 Modular Krylov construction 7
4.2 The autocorrelation function and Lanczos asymptotics 7
4.3 Proposed Krylov complexity growth rate 8

5 The Conjecture and Supporting Arguments **8**
5.1 Supporting arguments 9
5.2 The principal obstruction: free field vs. chaos 9

6 Open Questions and Research Programme **10**

1 Background and Motivation

1.1 Krylov complexity and the MSS chaos bound

Parker, Cao, Avdoshkin, Scaffidi, and Altman [2] introduced *Krylov* (Lanczos) complexity $\mathcal{C}_K(t)$ as a basis-independent measure of operator growth. Given an operator $\mathcal{O}$ evolving under a Hamiltonian H, one constructs iteratively the orthonormal *Krylov basis* $\{|O_n\rangle\}$ from the nested commutators $\mathcal{L}^n\mathcal{O}$ via the Gram–Schmidt procedure, where $\mathcal{L} = [H, \cdot]$ is the Liouvillian superoperator. The Krylov complexity is the mean position in this basis: $\mathcal{C}_K(t) = \sum_n n\,|\langle O_n|\mathcal{O}(t)\rangle|^2$. The key quantity in the recursion is the sequence of *Lanczos coefficients* $\{b_n\}_{n\geq 1}$, defined by the three-term recurrence.

Universal operator growth hypothesis [2]. *In generic (non-integrable) many-body systems, the Lanczos coefficients b_n grow asymptotically linearly:*

$$b_n \;\sim\; \alpha\, n \;+\; (\text{sub-leading}), \qquad n \to \infty, \tag{1}$$

where $\alpha > 0$ is a model-dependent rate, with a logarithmic correction in 1+1 dimensions. The rate α upper-bounds the Lyapunov exponent: $\lambda_L \leq 2\alpha$. Since the universal low-temperature bound gives $\lambda_L \leq 2\pi T = 2\pi/\beta$ (MSS), saturation of both bounds requires $\alpha = \pi/\beta$ and $b_n \sim (\pi/\beta)\, n$ in maximally chaotic systems.

The MSS chaos bound [3] states that in any large-N quantum system with a thermal density matrix at inverse temperature β, the OTOC Lyapunov exponent satisfies

$$\lambda_L \;\leq\; \frac{2\pi}{\beta}. \tag{2}$$

Equality holds in maximally chaotic systems including near-extremal black holes and the SYK model. The modular Krylov complexity version of MSS saturation, established numerically in SYK at large N [2], is $\lambda_K = 2\pi/\beta$ at saturation.

1.2 Heller–Papalini–Schuhmann: DSSYK and holographic Krylov

The paper by Heller, Papalini, and Schuhmann [1] (arXiv:2412.17785, PRL 135 151602, 2025) establishes a quantitative holographic match:

Verbatim abstract of [1]: "One of the important open problems in quantum black hole physics is a dual interpretation of holographic complexity proposals. To date the only quantitative match is the equality between the Krylov spread complexity in triple-scaled SYK at infinite temperature and the complexity = volume proposal in classical JT gravity. Our work utilizes the recent connection between double-scaled SYK and sine-dilaton gravity to show that the quantitative relation between Krylov spread complexity and complexity = volume extends to finite temperatures and to full quantum regime on the gravity side at disk level. From the latter we isolate the first quantum correction to the complexity = volume proposal and propose to view it as a complexity of quantum fields in the bulk. Finally, we comment on the switchback effect, whose presence would make the Krylov spread complexity a fully fledged holographic complexity at least in sine-dilaton gravity."

Remark 1 (Scope of HPS result)**.** The HPS result [1] is a *holographic* result about DSSYK/sine-dilaton gravity, not a derivation from algebraic quantum field theory or type-II crossed-product formalism. It establishes: (a) a quantitative $C = V$ match at finite temperatures in DSSYK; (b) the first quantum correction to $C = V$ as bulk-field complexity; (c) evidence for the switchback effect in DSSYK. It does *not* directly prove the Krylov exponent in a type-II$_\infty$ algebraic setting. The extension to FRW proposed in this note therefore requires additional algebraic input (the type-II$_\infty$ classification of the FRW diamond, §3) and an explicit conjecture [CONJECTURE] connecting HPS's holographic Krylov to the algebraic FRW modular flow.

1.3 Modular shadow and MSS saturation (ECI sec:mss-shadow)

The ECI programme [7] identifies the MSS chaos bound as a "modular shadow" of the saturation of the generalised entropy growth rate in the type-II$_\infty$ algebra. The ECI saturation identity is (eq. (20) of [7]):

$$\frac{1}{S_{\text{gen}}}\frac{dS_{\text{gen}}}{d\tau}\bigg|_{\text{sat}} = \frac{2\pi T_H}{\hbar} = \frac{\kappa_R}{\hbar}, \tag{3}$$

where $\kappa_R = 2\pi T_H$ is the Bisognano–Wichmann surface gravity. Under the Faulkner–Speranza identification $S_{\text{gen}} \leftrightarrow \mathcal{C}_K^{\text{mod}}$ [10], equation (3) produces the MSS bound at equality. **This identification is a working conjecture, not a theorem.** (ECI v6.0.44, §sec:mss-shadow, explicitly states: "We flag this identification as a working conjecture rather than a theorem: the bridge $S_{\text{gen}} \leftrightarrow \mathcal{C}_K^{\text{mod}}$ is consistent with the Witten–CLPW–Caputa–Faulkner–Speranza programme but has not been derived as a theorem in type-II$_\infty$ crossed-product algebras.")

The companion paper [6] develops this into the Modular Shadow Conjecture. The present note asks: does the same structural picture hold in the *FRW diamond*, where no global Killing vector exists?

2 The Heller–Papalini–Schuhmann Setting and Its Scope

2.1 DSSYK and sine-dilaton gravity

Heller, Papalini, and Schuhmann [1] work in the setting of *double-scaled SYK* (DSSYK), a variant of the Sachdev–Ye–Kitaev model defined by the limit $N, q \to \infty$ with $\lambda = 2\log q/N$ fixed. In this regime the model is dual, via the chord-diagram expansion, to *sine-dilaton gravity* on a two-dimensional background. The Krylov spread complexity in DSSYK at inverse temperature β is given by:

$$2|\log q|\,\mathcal{C}_K(t)_\beta \;=\; \langle \hat{L} \rangle \;=\; \left[-\partial_\Delta \langle e^{-\Delta \hat{L}} \rangle\right]_{\Delta=0}, \tag{4}$$

where $\hat{L}$ is the chord-length operator in sine-dilaton gravity. The central result of [1] is that $\mathcal{C}_K(t)_\beta = \langle \hat{L} \rangle_\beta / (2|\log q|)$ matches the complexity-equals-volume (C=V) proposal in sine-dilaton gravity at finite temperatures and at disk-level in the quantum $1/C_\phi$ expansion.

2.2 What is AdS/static-specific vs. what can transport to FRW

The HPS analysis relies on:

1. **Static patch / thermal state.** DSSYK at temperature β^{-1} corresponds to a static thermal state on the JT/sine-dilaton gravity side. There is an effective time-translation symmetry (the Hamiltonian of the SYK model) generating the thermal KMS flow.

2. **Chord diagram / matrix model structure.** The quantitative $C = V$ match uses chord combinatorics specific to DSSYK. These are not available in a general curved background.

3. **Holographic dictionary (JT/DSSYK).** The gravity dual is two-dimensional and its complexity = volume proposal is specific to dilaton gravity theories.

What survives in FRW: The *structural* content of the HPS result — that Krylov spread complexity is a holographically computable measure of operator growth that saturates the MSS bound at $\lambda_K = 2\pi/\beta$ — is what we wish to transport to the FRW diamond. The transport is possible because:

- The FRW diamond algebra $\mathcal{A}(\mathcal{O})_{\text{FRW}}$ is type-II$_\infty$ (established, §3), the same algebraic class as the AdS/JT algebra at large N.

- The KMS state on the FRW diamond at $\beta = 2\pi$ is the conformal pullback of the Minkowski vacuum (Hislop–Longo), which is structurally analogous to the Hartle–Hawking state in AdS.

- The Lanczos recursion is well-defined in any von Neumann algebra equipped with a faithful normal state (the modular Krylov construction of Caputa et al. [11]).

What does not transport:

- The quantitative $C = V$ match depends on the chord-diagram structure and the holographic dictionary of DSSYK. No FRW analogue of this dictionary exists.

- The quantum correction (bulk-field complexity) isolated by HPS is JT-gravity specific; in the FRW setting one would need a two- dimensional gravity dual for the FRW diamond, which is not available.
- There is no FRW analogue of the global Hamiltonian H_{SYK} generating the thermal flow; the modular Hamiltonian K_{FRW} (a non-local operator) must replace it.

3 FRW Diamond Setup

3.1 The metric and the comoving observer

We work in spatially flat ($k = 0$) FRW spacetime in conformal time η. The Hadamard property of the SLE state on the BKL attractor for Bianchi IX (a related non-stationary setting) is established in [15]; the type-II$_\infty$ analogy is an instructive comparison.

$$ds^2_{\mathrm{FRW}} \;=\; a(\eta)^2\big(-d\eta^2 + d\mathbf{x}^2\big), \qquad \mathbf{x} \in \mathbb{R}^3. \tag{5}$$

For radiation-dominated matter content ($p = \rho/3$), the Friedmann equations give $a(\eta) = a_0\,\eta$; for matter-dominated content ($p = 0$), $a(\eta) = a_0\,\eta^2$. Both cases are conformally flat, with the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ as conformal target.

A comoving observer at $\mathbf{x} = 0$ on $\eta \in (\eta_i, \eta_f)$ (with $0 < \eta_i < \eta_f < \infty$) sweeps out a *causal diamond*

$$D(\mathcal{O}) \;=\; \big\{(\eta, \mathbf{x}) \;:\; |\mathbf{x}| < \eta_f - \eta,\; \eta > \eta_i + |\mathbf{x}|\big\}. \tag{6}$$

In the conformally identified Minkowski coordinates, $D(\mathcal{O})$ is the same set of coordinate points as a doubly-bounded Minkowski diamond $M_{\mathcal{O}}$.

No global Killing vector. Neither radiation-dominated nor matter-dominated FRW admits a global timelike Killing vector field along a comoving worldline. The CLPW construction [8, 9] and its extensions to Killing horizons [10] are not directly applicable.

3.2 Conformally coupled scalar and the type-II$_\infty$ theorem

We consider the conformally coupled massless scalar field:

$$S[\varphi] \;=\; -\tfrac{1}{2}\int d^4x\,\sqrt{-g}\Big[g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi \;-\; \tfrac{1}{6}\,R\,\varphi^2\Big], \qquad \xi = \tfrac{1}{6}. \tag{7}$$

Under the Weyl rescaling $\tilde{\varphi} := a(\eta)\,\varphi$, the field equation $(\Box_g - R/6)\varphi = 0$ maps exactly to $\Box_{\mathrm{M}}\tilde{\varphi} = 0$ (conformal-covariance identity, sympy-verified in [4]). The Bunch–Davies (conformal) vacuum Ω_{FRW} maps to the Minkowski vacuum Ω_{M} under the unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implementing $\varphi(f) \mapsto \tilde{\varphi}(a^3 \cdot f)$.

The following theorem is established in [4]:

Theorem 2 (Type II$_\infty$ for the FRW diamond, [4] Theorem 1)**.** *Let φ be the conformally coupled massless scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond for a comoving observer with $0 < \eta_i < \eta_f < \infty$. Let σ^{FRW} denote the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$. Then:*

(1) $\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}(U^{-1}) \circ \sigma_t^{\mathrm{HL}} \circ \mathrm{Ad}(U)$*, where σ^{HL} is the Hislop–Longo modular automorphism group of the conformally identified Minkowski diamond.*

(2) The Connes spectrum satisfies $\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$.

(3) The crossed product $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ *is a type* II_∞ *factor.*

The same conclusion holds for matter-dominated FRW ($a(\eta) = a_0\eta^2$) *by the same proof.*

Remark 3 (No Killing vector required)**.** The proof of Theorem 2 uses only: (a) conformal flatness of g_{FRW}; (b) conformal invariance of the field equation ($\xi = 1/6$, $m = 0$); (c) the Hislop–Longo theorem on the conformally identified Minkowski diamond; (d) Tomita–Takesaki uniqueness (modular operators intertwine under the unitary U); (e) the Connes–Takesaki classification (Connes spectrum $\mathbb{R}$ implies type II_∞ of the crossed product). No timelike Killing vector of g_{FRW} is invoked. This is the first example of the CLPW type-II_∞ classification in a genuinely non-stationary FRW background with no Killing symmetry.

3.3 The FRW modular Hamiltonian and its KMS state

By Theorem 2(1), the FRW modular Hamiltonian is

$$K_{\mathrm{FRW}} \;=\; U^{-1}\, K_{\mathrm{HL}}\, U, \tag{8}$$

where $K_{\mathrm{HL}} = -\log \Delta_{\mathrm{HL}}$ is the Hislop–Longo modular Hamiltonian for the Minkowski diamond. The Hislop–Longo modular flow is the one-parameter Möbius group of conformal transformations of the Minkowski diamond fixing its tips; its generator K_{HL} is conjugate, via the conformal group $\mathrm{SO}(2,4)$, to a boost generator and has continuous spectrum filling $\mathbb{R}$. Consequently:

- K_{FRW} has spectrum $\mathrm{Spec}(K_{\mathrm{FRW}}) = \mathbb{R}$ (inherited from K_{HL} by unitary conjugation).
- The modular automorphism group $\sigma_s^{\mathrm{FRW}} = e^{isK_{\mathrm{FRW}}}(\cdot)e^{-isK_{\mathrm{FRW}}}$ implements a flow on $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ that corresponds, in the Minkowski picture, to the Möbius flow of the diamond, dressed by the conformal factor $a(\eta(s))/a(\eta)$ (explicit formula in [5]).
- The state Ω_{FRW} is a KMS state at inverse temperature $\beta = 2\pi$ with respect to the modular flow σ_s^{FRW}, by Tomita–Takesaki theory.

The conformal weight factor $a(\eta(s))/a(\eta)$ is the key difference from the static/AdS case. On the diamond orbit $(\eta(s), x^1(s)) = (\eta\cosh(2\pi s) + x^1\sinh(2\pi s),\ \eta\sinh(2\pi s) + x^1\cosh(2\pi s))$, this weight is:

$$w_{\mathrm{FRW}}(s;\eta,x^1) \;:=\; \frac{a(\eta(s))}{a(\eta)} \;=\; \frac{\eta\cosh(2\pi s) + x^1\sinh(2\pi s)}{\eta} \quad (a = a_0\eta), \tag{9}$$

which satisfies $w_{\mathrm{FRW}}(0;\eta,x^1) = 1$ and $w_{\mathrm{FRW}}(s) \to e^{2\pi s}$ as $s \to +\infty$ (on the wedge limit $x^1 \ll \eta$), recovering the Bisognano–Wichmann exponential at large boost parameter.

3.4 Scope and honest limitations

The following limitations of the FRW setup must be stated explicitly:

1. **Conformally coupled massless scalar only.** Theorem 2 applies to $\xi = 1/6$, $m = 0$. For $m > 0$ or $\xi \neq 1/6$, the field equation is not conformally covariant; the unitary U does not intertwine the algebras; the type of the crossed product is unknown.

2. **Bounded-away-from-singularity assumption.** The condition $\eta_i > 0$ (diamond bounded away from the Big Bang) is essential. When $\eta_i = 0$, the conformal rescaling $f \mapsto a^3 f$ degenerates and the isomorphism breaks.

3. **No quantitative $C = V$ dictionary.** Unlike the DSSYK/JT setting of [1], there is no known gravity dual for the FRW diamond that would give a quantitative complexity-equals-volume formula. All computations below are at the level of the modular Lanczos recursion, without a holographic check.

4. **ECI v6.0.44 explicit caveat.** Section 8 of the structural limitations of ECI v6.0.44 [7] states: "The type of $\mathcal{A}(\mathcal{O}) \rtimes_\sigma \mathbb{R}$ for $\mathcal{O}$ a causal patch of a non-stationary FRW observer (radiation- or matter-dominated era) is, to our knowledge, not derived in the published math.OA literature." The gap is filled by [4], submitted 2026-05-02, but has not yet been refereed. We proceed on the basis of the technical arguments of [4] while acknowledging pre-refereeing status.

4 Krylov Complexity in the FRW Diamond

4.1 Modular Krylov construction

The Krylov complexity of an operator $\mathcal{O}$ under the modular flow σ_s^{FRW} is defined via the modular Liouvillian $\mathcal{L}_{\mathrm{FRW}} = [K_{\mathrm{FRW}}, \cdot]$. The Krylov basis is generated by the Gram–Schmidt orthonormalisation of $\{\mathcal{L}_{\mathrm{FRW}}^n \mathcal{O}\}_{n\geq 0}$ with respect to the KMS inner product

$$(\mathcal{O}_1, \mathcal{O}_2)_\beta \;:=\; \int_0^\beta \frac{d\tau}{\beta} \langle \mathcal{O}_1^\dagger \, e^{-(\beta-\tau)K_{\mathrm{FRW}}} \, \mathcal{O}_2 \, e^{-\tau K_{\mathrm{FRW}}} \rangle_{\mathrm{FRW}}, \tag{10}$$

where $\langle\cdot\rangle_{\mathrm{FRW}}$ is the vacuum expectation value in the Bunch–Davies state. The Lanczos coefficients $\{b_n^{\mathrm{FRW}}\}$ are defined by the three-term recurrence in this inner product.

4.2 The autocorrelation function and Lanczos asymptotics

The Lanczos coefficients are determined by the modular autocorrelation function

$$C^{\mathrm{FRW}}(s) \;:=\; (\mathcal{O},\, \sigma_s^{\mathrm{FRW}}(\mathcal{O}))_\beta \;=\; \langle \mathcal{O}^\dagger \, \sigma_s^{\mathrm{FRW}}(\mathcal{O}) \rangle_{\mathrm{FRW}}. \tag{11}$$

By Theorem 2(1), this equals

$$C^{\mathrm{FRW}}(s) \;=\; \langle U\mathcal{O}U^{-1},\, \sigma_s^{\mathrm{HL}}(U\mathcal{O}U^{-1}) \rangle_{\mathrm{M}} \;=\; C^{\mathrm{HL}}(s), \tag{12}$$

where $C^{\mathrm{HL}}(s)$ is the Hislop–Longo autocorrelation function for the operator $\tilde{\mathcal{O}} := U\mathcal{O}U^{-1}$ in the Minkowski diamond vacuum. In other words, *the modular autocorrelation function in FRW is identical to the Hislop–Longo autocorrelation function in Minkowski, evaluated on the conformally related operator.*

For the free massless scalar, the Hislop–Longo two-point function on the Minkowski diamond at $\beta = 2\pi$ decays as

$$C^{\mathrm{HL}}(s) \;\sim\; \frac{1}{(2\sinh(\pi s))^{d-2}} \;\xrightarrow{d=4}\; \frac{1}{(2\sinh(\pi s))^2} \;\sim\; e^{-2\pi|s|} \quad (|s| \to \infty), \tag{13}$$

giving $C^{\mathrm{FRW}}(s) = C^{\mathrm{HL}}(s)$ with the same exponential decay. By the universal relationship between the spectral density of $C(s)$ and the Lanczos asymptotics (Parker et al. [2]), an

exponential decay $C(s) \sim e^{-2\pi|s|}$ corresponds to a spectral density supported on $[-2\pi, 2\pi]$ (an arc), giving linear Lanczos growth:

$$b_n^{\mathrm{FRW}} \sim \frac{\pi}{\beta} n = \frac{\pi}{2\pi} n = \frac{n}{2} \quad (\beta = 2\pi). \tag{14}$$

Remark 4 (Caution: free field vs. interacting)**.** The estimate (13) and its consequence (14) are for the *free* massless scalar, which is *not* maximally chaotic in the SYK sense. In free field theories, the Lanczos coefficients typically grow as $b_n \sim n^{1/2}$ (not linearly), leading to polynomial rather than exponential Krylov complexity growth. [IMPORTANT CAVEAT: the free-field autocorrelation decay rate does not generically imply linear Lanczos growth; the relationship between spectral density support and Lanczos asymptotics is more subtle. Parker et al. [2] show linear Lanczos growth for interacting and chaotic systems; for free fields the situation is different. The estimate (14) should be regarded as a formal analogy, not a rigorous derivation.] The step from $C^{\mathrm{FRW}}(s) = C^{\mathrm{HL}}(s)$ to $\lambda_K^{\mathrm{FRW}} = 2\pi/\beta$ requires either: (a) passing to an interacting field in the FRW diamond (breaking exact conformal invariance), or (b) invoking the operator growth hypothesis as applying to the modular flow on the type-II_∞ algebra in a "chaotic" sense to be made precise. This is the principal open gap.

4.3 Proposed Krylov complexity growth rate

Accepting the formal analogy and the modular Krylov construction of Caputa et al. [11], the Krylov complexity of an operator $\mathcal{O}$ under the FRW modular flow grows as

$$\mathcal{C}_K^{\mathrm{FRW}}(s) \sim e^{\lambda_K^{\mathrm{FRW}}|s|}, \qquad |s| \to \infty, \tag{15}$$

with the Krylov Lyapunov exponent

$$\lambda_K^{\mathrm{FRW}} = 2b_1^{\mathrm{FRW}} \approx \frac{2\pi}{\beta} = \frac{2\pi}{2\pi} = 1, \qquad (\beta = 2\pi). \tag{16}$$

This is the same rate as in the Rindler wedge [6] and in the SYK model [1,2], and saturates the MSS bound (2) exactly.

5 The Conjecture and Supporting Arguments

Conjecture 5 (FRW Krylov Saturation)**.** [CONJECTURE] *Let the setup be as in Theorem 2: conformally coupled massless scalar on radiation- or matter-dominated FRW, in the conformal vacuum, on a causal diamond bounded away from the Big Bang ($\eta_i > 0$). Then the Krylov complexity of operators evolved by the FRW modular flow σ_s^{FRW} grows exponentially at late modular times with Lyapunov exponent*

$$\lambda_K^{\mathrm{FRW}} = \frac{2\pi}{\beta_{\mathrm{FRW}}} = 1, \tag{17}$$

where $\beta_{\mathrm{FRW}} = 2\pi$ is the KMS inverse temperature of the Bunch–Davies vacuum with respect to the FRW modular flow. This saturates the MSS chaos bound (2) *exactly.*

5.1 Supporting arguments

A1. Algebraic universality. By Theorem 2, the FRW diamond algebra is type II_∞ with Connes spectrum $\Gamma = \mathbb{R}$ and modular flow isomorphic (via U) to the Hislop–Longo flow. The companion paper [6] conjectures that MSS saturation is a structural property of type-II_∞ algebras at KMS temperature $\beta = 2\pi$: the Bisognano–Wichmann surface gravity $\kappa_R = 2\pi T_H$ appears as a structural constant fixed by the algebraic data. If this conjecture is correct, Conjecture 5 follows as an instance in the FRW setting.

A2. Autocorrelation function argument. By (12), $C^{\mathrm{FRW}}(s) = C^{\mathrm{HL}}(s)$. The Hislop–Longo autocorrelation decays as $e^{-2\pi|s|}$ in $d = 4$ (at $\beta = 2\pi$), which, if one applies the Parker et al. universal operator growth hypothesis [2] to the modular flow, would give $b_n \sim n/2$ and $\lambda_K = 2\pi/\beta = 1$. The main gap (Remark 4) is that this argument requires the interacting/chaotic version of the growth hypothesis to apply to modular dynamics.

A3. Consistency with the DSSYK holographic result. Heller, Papalini, and Schuhmann [1] find $\lambda_K = 2\pi/\beta$ in DSSYK at finite temperature. The DSSYK model at large N (in the triple-scaled limit) approaches a type-II_∞ structure, and the KMS temperature of the Hartle–Hawking state in the gravity dual is identified with the SYK inverse temperature β. If Conjecture 5 is correct, it provides an independent (non-holographic) derivation of the same rate in a different type-II_∞ algebra — the FRW diamond — giving universal evidence that $\lambda_K = 2\pi/\beta$ is an algebraic invariant, not a holographic artefact.

A4. Modular shadow identification. From (3) and the Faulkner–Speranza identification $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$ (flagged as a conjecture), the ECI saturation rate $2\pi T_H$ equals $2\pi/\beta$ at $T_H = 1/(2\pi)$ (in units $k_B = \hbar = 1$, $\beta = 2\pi$). This is exactly the rate (17) in Conjecture 5. The FRW diamond realization provides a new geometrically distinct instance of this universal rate, strengthening the evidence for the modular shadow identification.

5.2 The principal obstruction: free field vs. chaos

The main obstruction to proving Conjecture 5 is the free-field nature of the setup. For free fields in Minkowski space, the modular Krylov complexity under the Bisognano–Wichmann boost flow does *not* generically exhibit exponential growth with $\lambda_K = 2\pi/\beta$; the free-field Lanczos coefficients grow sub-linearly and Krylov complexity grows polynomially or as a power law [2]. The exponential Krylov growth at $\lambda_K = 2\pi/\beta$ is a property of *maximally chaotic* interacting systems (SYK, near-extremal black holes), not of free fields.

The Araki relative modular operator approach to the $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$ bridge in type-II_∞ algebras is pursued in [16].

Three routes to resolve this:

(R1) **Interacting FRW fields.** Introduce a $\lambda\phi^4$ interaction on the FRW background. The conformally coupled scalar with $\lambda\phi^4$ is not conformally invariant in 4d (the β-function is non-trivial). However, the *type* of the crossed product (type II_∞) is expected to persist under weak interactions (it is a topological property of the modular flow), and the Lanczos growth rate might approach π/β in the strongly interacting (renormalized) theory. This requires an FRW analogue of the SYK large-N argument.

(R2) **Large-N limit in FRW.** Consider N copies of the conformally coupled scalar on the FRW diamond, with all-to-all ϕ^4-type couplings (SYK-like random coupling). In the large-N limit, one expects (by analogy with SYK [14]) a type-II_∞ modular flow with $\lambda_K \to 2\pi/\beta$. This is the most direct route and the most computationally tractable.

(R3) **Algebraic definition of "chaotic" modular flow.** Define a notion of "maximally chaotic modular automorphism" in a type-II_∞ algebra (e.g., via the Connes entropy or relative modular growth), and show that the Hislop–Longo flow, when restricted to an appropriate sub-algebra, falls in this class. This is the most mathematically ambitious route and the natural direction for the companion modular shadow paper [6].

6 Open Questions and Research Programme

Open Question 6. Does the modular Krylov complexity of the *free* conformally coupled scalar on the FRW diamond grow exponentially or polynomially? If polynomially, at what power? (This is an explicit computation using [2] applied to $C^{\mathrm{HL}}(s) \sim (2\sinh(\pi s))^{-2}$.)

Open Question 7. Do N conformally coupled scalars on the FRW diamond with large-N SYK-type random couplings satisfy $\lambda_K \to 2\pi/\beta$ as $N \to \infty$? What is the $O(1/N)$ correction to λ_K? (This mirrors the SYK analysis of [2] and the quantum correction of [1] in the holographic setting.)

Open Question 8. Does the same conjecture hold for the de Sitter static patch? In the static patch, the KMS temperature is $\beta_{\mathrm{dS}} = 2\pi/H$ (Gibbons–Hawking), giving $\lambda_K^{\mathrm{dS}} = H$ (the Hubble rate). The modular shadow conjecture then predicts $\lambda_K = H$, matching the de Sitter "scrambling rate" and connecting to cosmic censorship.

Open Question 9. Is there a two-dimensional gravity dual for the FRW diamond that would give a $C = V$ formula analogous to HPS [1]? The "FRW/CFT" correspondence (if it exists) would provide the quantitative check. A candidate is the celestial holography on the FRW past light cone, but no quantitative $C = V$ formula is known.

Open Question 10. What happens to λ_K^{FRW} as $\eta_i \to 0$? The Big Bang singularity is expected to introduce new structure (possibly a transition from type II_∞ to type II_1 or type III, mirroring the singularity theorem in the modular-flow setting). If λ_K diverges as $\eta_i \to 0$, this could be a quantum-gravity signature of the Big Bang.

Open Question 11. Is λ_K^{FRW} observable in the FRW setting? A possible route: analog gravity experiments simulating conformally coupled fields in a FRW-like background (e.g., acoustic Hawking radiation in an expanding BEC, cf. Steinhauer 2019 cited in [7]). The ECI Bisognano–Wichmann window $u_{\max} = 2\pi$ constrains the temporal window in which modular chaos is observable.

Proposed computational programme (~ 4 weeks).

1. *Week 1.* Explicit computation of $C^{\mathrm{FRW}}(s)$ for the free massless scalar via (12) and the known Hislop–Longo two-point function; Lanczos coefficients b_n^{FRW} for $n = 1, \ldots, 50$ via numerical recursion. Test whether b_n^{FRW} grows linearly ($\lambda_K = 1$) or sub-linearly.

2. *Week 2.* $N = 2, 4, 8$ conformally coupled scalars on a finite FRW diamond with SYK-type random ϕ^4 couplings. Compute OTOC Lyapunov exponent and check $\lambda_L \leq 2\pi/\beta$. Does $\lambda_L \to 2\pi/\beta$ as N or coupling increases?

3. *Week 3.* Analytical approximation of b_n^{FRW} in the large-n limit using the spectral density of K_{FRW} via the Kac–Moody generator conjugation (8). Attempt asymptotic analysis using the known Möbius-flow spectral measure on the diamond.

4. *Week 4.* Write-up; connection to the de Sitter case (Question 8); comparison with DSSYK quantum corrections of [1]; submission as Comm. Math. Phys. paper (or companion preprint to [6]).

Acknowledgments

This note was drafted as Opus Bridge #16, extending the DSSYK/Krylov holographic complexity analysis of [1] to the FRW diamond, using the type-II_∞ classification of [4] and the modular shadow framework of [6, 7]. The [CONJECTURE] tags throughout reflect the current status of the central claim (Conjecture 5): supported by structural arguments but not proven.

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